

A Second-Order Memristor Method to Construct Memristive Neural Networks With Multi-Butterfly and Multi-Scroll Dynamics

Hairong Lin[✉], *Senior Member, IEEE*, Xiaoheng Deng[✉], *Senior Member, IEEE*, Yi Zhang[✉],
and Geyong Min[✉], *Member, IEEE*

Abstract—Memristor-based Hopfield neural networks (MHNNs) exhibit rich chaotic dynamics and bear closer hardware resemblance to the biological brain, making them well suited for emulating neural dynamical behaviors. However, most existing MHNNs are constructed with first-order memristors. This paper proposes a novel second-order memristor (SOM) approach for constructing MHNNs with enriched chaotic dynamics. Specifically, a second-order memristor is incorporated into a three-neuron Hopfield neural network to emulate the magnetic coupling mechanism between neurons, thereby forming a second-order memristor-based neural network (SOM-HNN). Comprehensive dynamical analyses, including bifurcation diagrams, Lyapunov exponent spectra, and numerical simulations, confirm that the proposed SOM-HNN exhibits richer and more intricate chaos behaviors than its first-order counterparts. Remarkably, the proposed SOM-HNN can simultaneously generate butterfly and scroll attractors, multi-butterfly and multi-scroll attractors, as well as initial-boosed coexisting multi-butterfly and multi-scroll attractors, thereby substantially enhancing its dynamical diversity. To the best of our knowledge, this is the first report of both multi-butterfly and multi-scroll dynamics in a neural network. Furthermore, the SOM-HNN is implemented in hardware using analog circuits and a digital field-programmable gate array (FPGA) platform. Experimental results demonstrate the network's abundant dynamical features and its feasibility for efficient hardware realization in neuromorphic engineering applications.

Index Terms—Second-order memristor, bionic modeling, Hopfield neural network, multi-butterfly attractor, multi-scroll attractor, circuit implementation.

Received 11 November 2025; revised 28 December 2025; accepted 7 February 2026. This work was supported in part by the National Natural Science Foundation of Hunan Province under Grant 2026JJ50518, in part by the National Natural Science Foundation of China Project under Grant 62572494 and Grant 62172441, in part by the Joint Funds for Railway Fundamental Research of National Natural Science Foundation of China under Grant U2368201, in part by the Special Fund of National State Key Laboratory of Ni & Co Associated Minerals Resources Development and Comprehensive Utilization under Grant GZSYSKY-2024-073, in part by the Key Project of Shenzhen City Special Fund for Fundamental Research under Grant 202208183000751, and in part by the High Performance Computing Center of Central South University. This article was recommended by Associate Editor E. Donati. (Corresponding author: Xiaoheng Deng.)

Hairong Lin, Xiaoheng Deng, and Yi Zhang are with the School of Electronic Information, Central South University, Changsha 410083, China, and also with Shenzhen Research Institute, Central South University, Shenzhen 518000, China (e-mail: haironglin@csu.edu.cn; dxh@csu.edu.cn; 254512054@csu.edu.cn).

Geyong Min is with the Department of Computer Science, University of Exeter, EX4 4QF Exeter, U.K. (e-mail: g.min@exeter.ac.uk).

Digital Object Identifier 10.1109/TCSI.2026.3663432

I. INTRODUCTION

SINCE the first nanoscale memristor was reported by HP Labs in 2008 [1], extensive research efforts have been devoted to exploring memristor and its broad applications in nonlinear circuits and systems [2], [3], as illustrated in Fig. 1. Among these, a particularly active direction is the development of memristor-based Hopfield neural networks (MHNNs) [4], which leverage the distinctive properties of memristors. Owing to their intrinsic memory capability, strong nonlinearity, and nanoscale features, MHNNs exhibit a more faithful resemblance to biological neural systems compared with conventional neural networks, especially in modeling complex neurodynamics and enabling highly integrated hardware implementations [5], [6]. As a result, the construction and dynamical analysis of MHNNs with abundant and diverse behaviors have become a focal point of interest for researchers worldwide.

Over the past two decades, considerable efforts have been dedicated to constructing MHNNs capable of exhibiting diverse and complex dynamical behaviors. Broadly, existing research can be categorized into three main directions. (1) MHNNs with general chaotic dynamics such as hyperchaos [7], coexisting attractors [8], and hidden attractors [9]. For instance, Li et al. [10] developed a hyperchaotic MHNN by employing a memristor to emulate synaptic coupling, while Bao et al. [11] introduced a hyperbolic memristor-based HNN with coexisting asymmetric attractors. Ding et al. [12] further proposed an MHNN with hidden hyperchaotic attractors by modeling the magnetic induction effect with a memristor. Additional studies have reported various hyperchaotic, hidden, and coexisting attractors in different neural network frameworks [13], [14]. (2) Multi-scroll chaotic dynamics, including multi-scroll, grid multi-scroll, and space multi-scroll attractors [15], [16]. Zhang et al. [17] showed that an MHNN with a multi-piecewise quadratic memristor synapse can generate single-directional multi-scroll attractors. Lai et al. [18] revealed that adjustable memductance and dual memristors enable the formation of grid multi-scroll attractors, while Vignesh et al. [19] demonstrated that fractional-order MHNNs can also exhibit multi-scroll dynamics. Other forms, such as hidden, symmetric, and coexisting multi-scroll attractors, have also been reported [20], [21], [22]. (3) Multi-wing chaotic dynamics, including multi-wing attractors [23] and

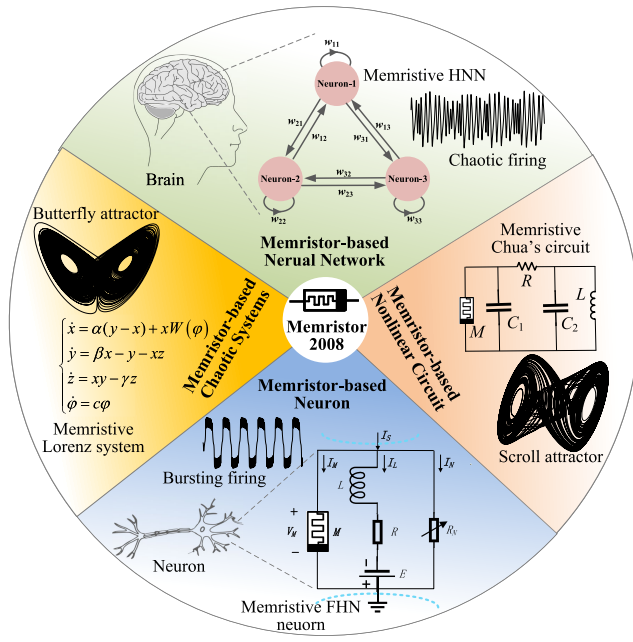


Fig. 1. Summary of the research on Memristor-based applications in nonlinear circuits and systems, including memristor-based neural networks, memristor-based nonlinear circuits, memristor-based neurons, and memristor-based chaotic systems. The most widely studied memristive neural network is memristive Hopfield neural network (HNN).

multi-butterfly attractors [24]. Compared with other dynamical forms, multi-wing attractors exhibit more intricate mechanisms and richer nonlinear features [25], [26]. For a long time, realizing such dynamics in MHNNs remained a challenge. Only recently, Lin et al. [27] achieved multi-butterfly attractors by employing different memristors to emulate distinct biological functions, followed by Deng et al. [28], who proposed a multi-wing MHNN via a multi-segment memristor coupled to two neurons. Nevertheless, constructing an MHNN that simultaneously produces both multi-wing and multi-scroll dynamics remains an open and challenging problem.

The above analysis highlights that memristors play a pivotal role in shaping the dynamical behaviors of MHNNs. Consequently, considerable efforts have been devoted to constructing novel MHNNs by incorporating different types of memristors. With the discovery of locally active memristors (LAMs) [29], [30], LAM-based HNNs have attracted increasing attention. Compared with conventional memristors, LAMs exhibit richer and more intricate dynamic characteristics, enabling MHNNs to generate unique neurodynamical behaviors. For example, Li et al. [31] developed an MHNN exhibiting complex bursting firing patterns by employing a tristable LAM to emulate synaptic coupling. Initial-boosted dynamics have also been reported in LAM-based HNNs [32]. In addition, hidden multi-scroll attractors [33] as well as homogeneous and heterogeneous coexisting attractors [33] have been revealed, further demonstrating the potential of LAMs in enhancing the dynamical richness of neural networks.

The second-order memristor (SOM), experimentally demonstrated in 2015 [34], represents a novel class of memristive devices. Unlike first-order and locally active memristors, the

SOM possesses two internal state variables, and its memductance depends on both states [35], [36]. This dual-state property allows it to simultaneously emulate two biological functions. Owing to its enhanced nonlinearity, SOM-based systems are capable of generating more intricate dynamical behaviors compared with those built on first-order memristors. In recent years, SOMs have been employed to construct diverse memristive systems with enriched dynamical complexity. For instance, Marrone et al. [37] proposed a simplified SOM model to design a memristive spiking network capable of performing multiple neural functions. Mou et al. [38] introduced a discrete SOM-based chaotic map and demonstrated its superior dynamical richness over first-order memristor-based maps. Mao et al. [39] provided a systematic summary of general SOM models and proposed a neural circuit operating near the edge of chaos. Furthermore, Mou et al. [40] designed a novel SOM model to construct a memristive Rulkov neuron with diverse neural dynamics, while Rodríguez-Martínez et al. [41] developed a coupled neuron model by employing an SOM to emulate synaptic connections. Despite these advances, SOM-based neural networks have not yet been systematically investigated.

Motivated by the above analysis, this work proposes a second-order memristor-based Hopfield neural network (SOM-HNN), in which a novel second-order memristor is employed to emulate the magnetic coupling mechanism between two neurons. The dynamical properties of the SOM-HNN are systematically investigated through comprehensive theoretical and numerical analyses. The results demonstrate that the SOM-HNN is capable of generating both multi-scroll and multi-butterfly attractors, with the number of scrolls and butterflies effectively controlled by the memristor parameters. In addition, the SOM-HNN exhibits complex initial-offset boosting phenomena, further enriching its dynamical repertoire. To validate the feasibility of the proposed model, both analog and digital circuit implementations are developed, and the experimental results are shown to be consistent with the numerical simulations.

The remainder of this article is organized as follows: a second-order memristor is designed in Section II. The model and dynamics analysis of SOM-HNN are given in Section III. Section IV describes the analog and digital hardware implementations. Finally, Section V concludes the work and discusses future research directions.

II. A MULTI-PIECEWISE SECOND-ORDER MEMRISTOR MODEL

In this section, a novel second-order memristor model is first proposed. Its voltage–current characteristics are then analyzed, followed by the design and validation of a circuit emulator.

A. Mathematical Model Design

According to Chua’s unfolding theorem and Ohm’s law [42], the generic voltage-controlled second-order memristor model can be expressed as follows:

Double-state-dependent ohm’s law:

$$i = W(\varphi_1, \varphi_2)v. \quad (1)$$

Double-state-controlled equations:

$$\begin{cases} \dot{\varphi}_1 = f_1(\varphi_1, \varphi_2, v), \\ \dot{\varphi}_2 = f_2(\varphi_1, \varphi_2, v). \end{cases} \quad (2)$$

Here, v and i denote the voltage and the current across the memristor, while φ_1 and φ_2 are its two internal state variables. The functions f_1 and f_2 describe the dual state-control equations, and W represents the memductance function adjusted by these two state variables. Based on (1) and (2), a new flux-controlled multi-piecewise second-order memristor model is designed as follows:

$$\begin{cases} i = W(\varphi_1, \varphi_2)v = (\alpha - \beta\varphi_1 - \gamma\varphi_2)v, \\ \dot{\varphi}_1 = a - bv^2 - c\varphi_1, \\ \dot{\varphi}_2 = pv - qh(\varphi_2), \end{cases} \quad (3)$$

where $\alpha, \beta, \gamma, a, b, c, p$ and q are positive memristor parameters. The function $h(\varphi_2)$ is described by

$$h(\varphi_2) = \begin{cases} \varphi_2, M < 0 \\ \varphi_2 + M - \sum_{i=0}^M \text{sgn}(\varphi_2 + 2i), M = 0, 1, 2, \dots \end{cases} \quad (4)$$

where M is tunable control parameter. Consequently, unlike existing second-order memristor models [37], [38], [39], [40], [41], the proposed second-order memristor offers enhanced tunability and flexibility, making it well suited to induce a diversity range of dynamical behaviors in memristive systems.

Next, the boundedness of the two internal state variables φ_1 and φ_2 is investigated. First, by integrating both sides of the first state equation in Eq. (3), we obtain

$$c\varphi_1^2/2 - (a - bv^2 - 1)\varphi_1 - P = 0, \quad (5)$$

where P is an integration constant. Further solving Eq. (5) yields

$$\varphi_1 = (a - bv^2 - 1) \pm \frac{\sqrt{(a - bv^2 - 1)^2 + 2cP}}{c}. \quad (6)$$

Since the parameters a, b , and c are positive constants, and assuming that the input voltage v is a bounded signal satisfying $(-V_T \leq v \leq V_T)$, where V_T denotes a given threshold, the bound of the state variable φ_1 can be derived as

$$\begin{cases} \varphi_1 \geq (a - bV_T^2 - 1) - \frac{\sqrt{(a - bV_T^2 - 1)^2 + 2cP}}{c}, \\ \varphi_1 \leq (a - 1) + \frac{\sqrt{(a - 1)^2 + 2cP}}{c}. \end{cases} \quad (7)$$

It is evident that under any bounded input voltage v ($-V_T \leq v \leq V_T$), the state variable φ_1 remains bounded.

Next, the boundedness of φ_2 is analyzed. By integrating both sides of the second state equation in Eq. (3), we obtain

$$q\varphi_2^2/2 + (qM - pv + 1)\varphi_2 - q \sum_{i=0}^M |\varphi_2 + 2i| + Q = 0, \quad (8)$$

where Q is an integration constant. The solution of Eq. (8) depends on the sign of $(\varphi_2 + 2i)$, which leads to the following two cases.

Case 1: $(\varphi_2 - i) \geq 0$.

In this case, Eq. (8) can be rewritten as

$$q\varphi_2^2/2 + (1 - pv - q)\varphi_2 - qM(M + 1) + Q = 0, \quad (9)$$

and its solution is given by

$$\varphi_2 = -(1 - pv - q) \pm \frac{\sqrt{(1 - pv - q)^2 - 2q(Q - qM(M + 1))}}{q}. \quad (10)$$

Since the parameters p , and q are positive constants, and assuming that the input voltage v ($-V_T \leq v \leq V_T$) is bounded, the bound of the state variable φ_2 can be obtained as

$$\begin{cases} \varphi_2 \geq -(1 + pV_T - q) \pm \frac{\sqrt{(1 + pV_T - q)^2 - 2q(Q - qM(M + 1))}}{q}, \\ \varphi_2 \leq -(1 - pV_T - q) + \frac{\sqrt{(1 - pV_T - q)^2 - 2q(Q - qM(M + 1))}}{q}. \end{cases} \quad (11)$$

Case 2: $(\varphi_2 - i) < 0$.

Similarly, Eq. (8) can be transformed into

$$q\varphi_2^2/2 + (1 - pv + q + 2qM)\varphi_2 + qM(M + 1) + Q = 0, \quad (12)$$

whose solution is expressed as

$$\varphi_2 = -(1 - pv + q + 2qM) \pm \frac{\sqrt{(1 - pv + q + 2qM)^2 - 2q(Q + qM(M + 1))}}{q}. \quad (13)$$

Under the same bounded voltage excitation, the corresponding bound of φ_2 can be derived as

$$\begin{cases} \varphi_2 \geq -(1 - pV_T + q + 2qM) - \frac{\sqrt{(1 - pV_T + q + 2qM)^2 - 2q(Q + qM(M + 1))}}{q}, \\ \varphi_2 \leq -(1 + pV_T + q + 2qM) + \frac{\sqrt{(1 + pV_T + q + 2qM)^2 - 2q(Q + qM(M + 1))}}{q}. \end{cases} \quad (14)$$

From the above analysis, it follows that for any bounded input voltage v ($-V_T \leq v \leq V_T$), the state variable φ_2 is also bounded. Consequently, both internal state variables of the proposed second-order memristor exhibit bounded trajectories under bounded voltage excitation. This boundedness property is essential for ensuring the generation of physically realizable and bounded chaotic dynamics in memristive systems. Moreover, Eqs. (11) and (14) indicate that the control parameter M can flexibly regulate the extreme values of the second state variable. This physical property endows the memristive system with a high degree of tunability and randomness, thereby enabling the generation of richer and more complex chaotic dynamical behaviors.

B. Voltage-Current Characteristics

To verify the presented mathematical model corresponds to a second-order memristor device, its voltage-current relationship is analyzed. The model is studied under different signal frequencies and amplitudes, using a sinusoidal voltage source $v = A \sin(Ft)$, where A is the amplitude and F is the frequency. The memristor parameters are set as $\alpha = \beta = 1$, $\gamma = 0.01$, $a = 1$, $b = 1$, $c = 0.1$, $p = 2.2$, $q = 5$,

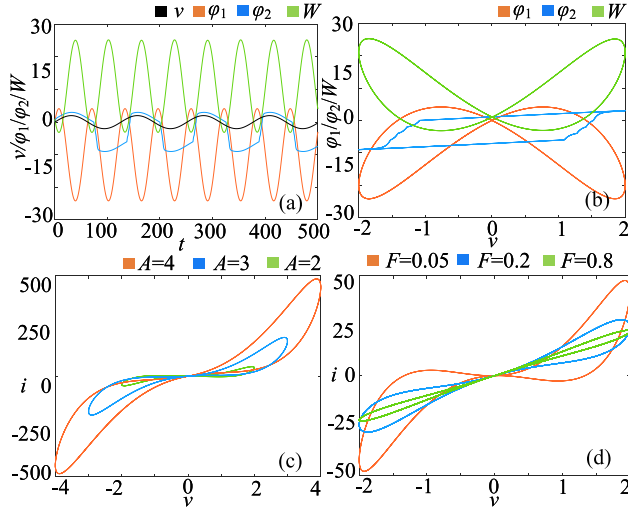


Fig. 2. Dynamical feature of the second-order memristor under $v = A\sin(Ft)$ with parameters $\alpha = \beta = 1, \gamma = 0.01, a = 1, b = 1, c = 0.1, p = 2.2, q = 5, M = 3$, initial values $(\phi_{10}, \phi_{20}) = (0, 0)$. (a) $v/\phi_1/\phi_2/W - t$ time series with $A = 2, F = 0.05$. (b) $\phi_1/\phi_2/W - v$ relationship with $A = 2, F = 0.05$. (c) A -dependent pinched hysteresis loops with $F = 0.05$. (d) F -dependent pinched hysteresis loops with $A = 2$.

$M = 3$, with initial values $(\phi_{10}, \phi_{20}) = (0, 0)$. When $A = 2$ and $F = 0.05$, the time responses of the state variables ϕ_1, ϕ_2 and W are shown in Fig. 2(a), exhibiting periodic and bounded behavior. As illustrated in Fig. 2(b), the state variables are directly controlled by the applied voltage, indicating that the memductance W varies with v . This confirms that the proposed model possesses dynamic memory and bounded behaviors. Further simulations are performed under different amplitudes ($A = 2, 3, 4$) with fixed frequency $F = 0.05$, and under different frequencies ($F = 0.05, 0.2, 0.8$) with fixed amplitude $A = 2$. The corresponding pinched hysteresis loops are depicted in Fig. 2(c) and (d), respectively. In all six cases, the loops pass through the origin in the voltage-current ($v - i$) plane, a hallmark of memristive behavior. Moreover, Fig. 2(d) shows that as the excitation frequency increases from 0.05 to 0.8, the hysteresis lobe area gradually decreases. When the frequency approaches infinity, the loop tends toward a single-valued function. Therefore, the proposed mathematical model satisfies the three fingerprints of a memristor [42], confirming that the system described by Eq. (3) is indeed a second-order memristor device.

C. Circuit Emulator Implementation

The circuit emulator of the proposed second-order memristor is implemented using an analog circuit. Following the design principles of nonlinear circuits, the mathematical model described in Eqs. (3)–(4) is transformed into the corresponding circuit state equations as follows.

$$\begin{cases} i = v/R_\alpha - g v v_{\phi 1}/R_\beta - g v v_{\phi 2}/R_\gamma, \\ C_1 \dot{v}_{\phi 1} = e/R_a - g v^2/R_b - v_{\phi 1}/R_c, \\ C_2 \dot{v}_{\phi 2} = v/R_p - v_h/R_q, \end{cases} \quad (15)$$

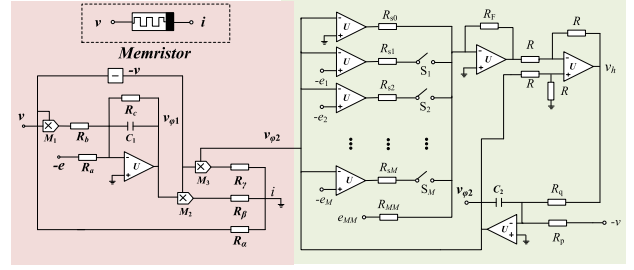


Fig. 3. Second-order memristor circuit emulator.

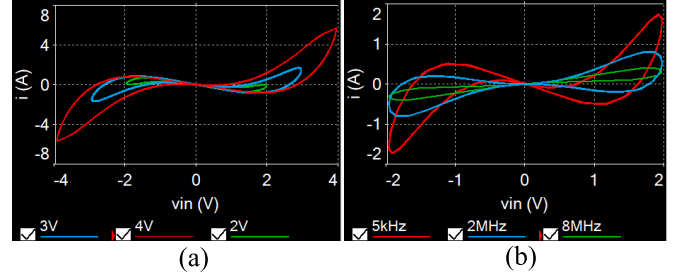


Fig. 4. Physical circuit experiment results of the second-order memristor. (a) H -dependent pinched hysteresis loops with $f = 5$ kHz. (b) f -dependent pinched hysteresis loops with $H = 2$ V.

where v and i are input voltage and output current, respectively, the voltages across the two capacitors (C_1 and C_2), denoted as $v_{\phi 1}$ and $v_{\phi 2}$, represent the two memristor state variables. Here, RC is the integration time constant, g is the coefficient of the analog multiplier, and R_a and R_b are passive resistors corresponding to memristor parameters. Based on the circuit Eq. (15), a second-order memristor circuit emulator is realized, as shown in Fig. 3. The circuit emulator consists of two function parts highlighted in red and green background: the red section implements the memductance function and the first memristor state equation, while the green section implements the second memristor state equation. In the green section, the control parameter M is configured by switching $S_1 - S_M$. The signals $e_1 - e_M$ and e_{MM} serve as the input control voltages.

The implemented second-order memristor circuit emulator is validated in Multisim using an input voltage of $v = H\sin(2\pi ft)$, where $H = A$ and $f = F/RC$ are the stimulus amplitude and frequency, respectively. Based on the parameter values employed in the numerical simulations shown in Fig. 2(c) and (d), the resistances in Fig. 3 are determined as follows

$$\begin{cases} R_\alpha = R/\alpha = 10\text{k}\Omega, R_\beta = gR/\beta = 1\text{k}\Omega, R_\gamma = gR/\gamma = 100\text{k}\Omega, \\ R_a = e/a = 1\text{k}\Omega, R_b = gR/b = 1\text{k}\Omega, R_c = R/c = 100\text{k}\Omega, \\ R_p = R/p = 4.5\text{k}\Omega, R_q = R/q = 2\text{k}\Omega, R = 10\text{k}\Omega, \\ R_{S1} = 13.5\text{k}\Omega, R_{MM} = R_F = 1\text{k}\Omega, C = 1\text{nF}. \end{cases} \quad (16)$$

When $RC = 10^{-5}$, $f = 5$ kHz is fixed and H is set to 2 V, to 3 V, and to 4 V, respectively, the $v - i$ curves are given in Fig. 4(a). While when $H = 2$ V is fixed and f is set to 5 kHz, to 2 MHz, and to 8 MHz, respectively, the $v - i$ curves are displayed in Fig. 4(b). As shown in Fig. 4, the circuit simulation results are highly consistent with the corresponding

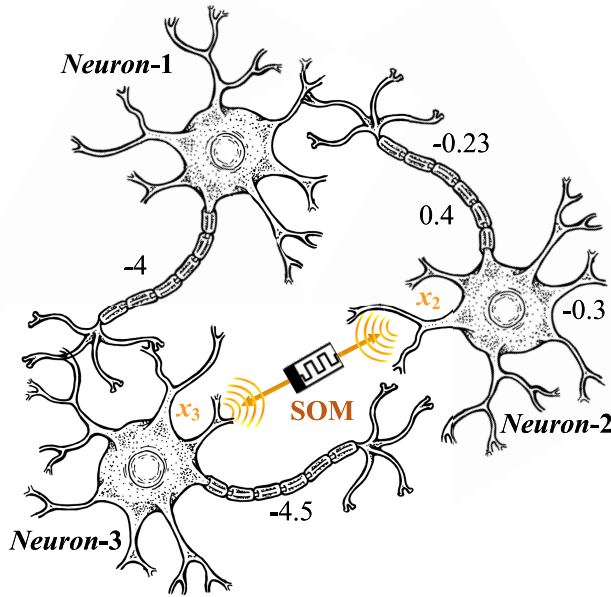


Fig. 5. Connection structure of the neural network with magnetic coupling behavior.

numerical simulation results in Fig. 2(c) and (d). This confirms that the implemented circuit simulator can indeed perform the function of the memristor.

III. SECOND-ORDER MEMRISTOR-BASED HOPFIELD NEURAL NETWORK

In this section, a second-order memristor-based neural network is constructed, and its equilibria stability, complex dynamical behaviors are analyzed through theoretical derivation and numerical simulations. All numerical experiments are carried out in MATLAB R2021a using the ODE23 solver. The initial time, integration step size, and total simulation duration are specified as 500, 0.01, and 3000, respectively. In addition, the Lyapunov exponents are evaluated according to Wolf's algorithm.

A. SOM-HNN Model Description

The Hopfield neural network (HNN) is an abstract model of the brain with rich chaotic dynamics and is commonly employed to investigate the dynamical behaviors of nervous system. The original HNN can be expressed as follows [43]:

$$C_i \dot{x}_i = -\frac{x_i}{R_i} + \sum_{j=1}^n w_{ij} \tanh(x_j) + I_i, \quad i, j \in N^* \quad (17)$$

Here, x_i denotes the membrane potential of neuron i , while C_i and R_i represent the corresponding membrane capacitance and resistance, respectively. The term w_{ij} stands for the coupling strength from neuron j to neuron i , namely, the synaptic weight. Additionally, $\tanh(\cdot)$ and I_i denote the neuron activation function and the external input current, respectively. For simplicity, the parameters are often normalized as $C_i = 1$, $R_i = 1$, and $I_i = 0$. Based on Eq. (17), a novel

small HNN comprising three neurons can be constructed as follows

$$\begin{cases} \dot{x}_1 = -x_1 + w_{11} \tanh(x_1) + w_{12} \tanh(x_2) + w_{13} \tanh(x_3), \\ \dot{x}_2 = -x_2 + w_{21} \tanh(x_1) + w_{22} \tanh(x_2) + w_{23} \tanh(x_3), \\ \dot{x}_3 = -x_3 + w_{31} \tanh(x_1) + w_{32} \tanh(x_2) + w_{33} \tanh(x_3), \end{cases} \quad (18)$$

where x_i ($i = 1, 2, 3$) denotes the membrane potentials of neurons i . For the constructed HNN in Eq. (18), the synaptic weight coefficients are set as $w_{11} = 0$, $w_{12} = 0.4$, $w_{13} = -4$, $w_{21} = -0.23$, $w_{22} = -0.3$, $w_{23} = 0$, $w_{31} = 0$, $w_{32} = -4.5$, and $w_{33} = 0$. Under this configuration, the proposed HNN only exhibits stable point behavior. However, when considering a magnetic coupling between neuron 2 and neuron 3 due to the potential difference across their membranes [44], the HNN will generate more complex dynamical behaviors. This coupling induces a magnetic current, which in turn modulates the neurons' membrane potentials and influences the overall network dynamics. Leveraging the flux-dependent property of memristors, this magnetic-induced current can be effectively modeled [45]. As illustrated in Fig. 5, by incorporating the designed second-order memristor to represent the magnetic coupling in the small HNN of Eq. (18), a second-order memristor-based neural network, namely SOM-HNN model is constructed as follows:

$$\begin{cases} \dot{x}_1 = -x_1 + 0.4 \tanh(x_2) - 4 \tanh(x_3), \\ \dot{x}_2 = -x_2 - 0.23 \tanh(x_1) - 0.3 \tanh(x_2) \\ \quad - kW(\varphi_1, \varphi_2)(x_2 - x_3), \\ \dot{x}_3 = -x_3 - 4.5 \tanh(x_2) + kW(\varphi_1, \varphi_2)(x_2 - x_3), \\ \dot{\varphi}_1 = a - b(x_2 - x_3)^2 - c\varphi_1, \\ \dot{\varphi}_2 = p(x_2 - x_3) - qh(\varphi_2). \end{cases} \quad (19)$$

Here, $kW(x_2 - x_3)$ denotes the electromagnetic induction current induced by the membrane potential difference ($x_2 - x_3$) between neuron 2 and neuron 3, where k represents the coupling intensity of the magnetic interaction. The memductance $W(\varphi_1, \varphi_2)$ characterizes the relationship between magnetic flux and the membrane potential difference. Variables φ_1 and φ_2 jointly describe the magnetic flux induced by the potential difference. The fourth and fifth equation in Eq. (19) capture the contribution of the membrane potential difference to the evolution of magnetic flux.

B. Equilibria Stability Analysis

The equilibrium points and their corresponding stabilities of the proposed SOM-HNN are investigated. By setting the left-hand side of system (19) equal to zero, the equilibrium points of the SOM-HNN can be numerically obtained. Assume that the system admits an equilibrium point $E = (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{\varphi}_1, \tilde{\varphi}_2)$, then, the equilibrium equations of the SOM-HNN can be formulated as follows.

$$\begin{cases} 0 = -\tilde{x}_1 + 0.4 \tanh(\tilde{x}_2) - 4 \tanh(\tilde{x}_3), \\ 0 = -\tilde{x}_2 - 0.23 \tanh(\tilde{x}_1) - 0.3 \tanh(\tilde{x}_2) \\ \quad - kW(\tilde{\varphi}_1, \tilde{\varphi}_2)(\tilde{x}_2 - \tilde{x}_3), \\ 0 = -\tilde{x}_3 - 4.5 \tanh(\tilde{x}_2) + kW(\tilde{\varphi}_1, \tilde{\varphi}_2)(\tilde{x}_2 - \tilde{x}_3), \\ 0 = a - b(\tilde{x}_2 - \tilde{x}_3)^2 - c\tilde{\varphi}_1, \\ 0 = p(\tilde{x}_2 - \tilde{x}_3) - qh(\tilde{\varphi}_2). \end{cases} \quad (20)$$

TABLE I
EQUILIBRIA, CORRESPONDING EIGENVALUES, AND THEIR STABILITIES UNDER DIFFERENT CASES

Cases	Equilibria	Eigenvalues	Stabilities
Case 1: $M < 0, a = 0$	(0, 0, 0, 0, 0)	(-5, -1.6932±1.7734i, -1.9136, -0.1)	Stable focus
Case 2: $M < 0, a \neq 0, a=1$	(0, 0, 0, 10, 0)	(-5, 18.8430, -3.3877, -0.7553, -0.1)	Unstable saddle
Case 3: $M \geq 0, M = 0, a = 0$	(0, 0, 0, 0, 1)	(-5, -1.6777±1.7696i, -1.9247, -0.1)	Stable focus
	(0, 0, 0, 0, -1)	(-5, -1.7088±1.7772i, -1.9025, -0.1)	Stable focus
Case 4: $M \geq 0, M = 1, a \neq 0, a = 1$	(0, 0, 0, 10, 1)	(-5, 18.8633, -3.3878, -0.7554, -0.1)	Unstable saddle
	(0, 0, 0, 10, -1)	(-5, 18.8228, -3.3875, -0.7552, -0.1)	Unstable saddle

Eq. (20) can be further simplified to the following equation

$$\begin{cases} \tilde{x}_1 = 0.4 \tanh(\tilde{x}_2) - 4 \tanh(\tilde{x}_3), \\ \tilde{x}_1 = \arctan h((-0.75 \tanh(\tilde{x}_2) - \tilde{x}_2 - \tilde{x}_3)/0.23), \\ \tilde{x}_3 = (kW(\tilde{\varphi}_1, \tilde{\varphi}_2)\tilde{x}_2 - 4.5 \tanh(\tilde{x}_2))/(1 + kW(\tilde{\varphi}_1, \tilde{\varphi}_2)), \\ \tilde{\varphi}_1 = (a - b(\tilde{x}_2 - \tilde{x}_3)^2)/c, \\ h(\tilde{\varphi}_2) = p(\tilde{x}_2 - \tilde{x}_3)/q. \end{cases} \quad (21)$$

Based on Eq. (21), the equilibrium points of the SOM-HNN can be classified into the following cases. Case 1: when $M < 0, a = 0$, and all other parameters are positive, the SOM-HNN admits a zero equilibrium point $E=(0, 0, 0, 0, 0)$. Case 2: when $M < 0, a \neq 0$, and all other parameters are positive, the system possesses a non-zero equilibrium point $E=(0, 0, 0, a/c, 0)$. Case 3: when $M \geq 0, a = 0$, and all other parameters are positive, the system exhibits infinitely many nonzero equilibrium points given by $E=(0, 0, 0, 0, r)$, where r is an integer satisfying $(-2M < r < 2)$. The total number of equilibrium points depends on the control parameter M and equals $(2M + 3)$. Case 4: when $M \geq 0, a \neq 0$, and all other parameters are positive, infinitely many non-zero equilibrium points arise as $E=(0, 0, 0, a/c, r)$, with r being an integer in the range $(-2M < r < 2)$. Similarly, the number of equilibrium points is determined by the control parameter M and is equal to $(2M + 3)$. Next, the stability characteristics of these equilibrium points are examined. For a dynamical system, stability analysis is typically carried out by linearizing the system around an equilibrium point, constructing the Jacobian matrix, and evaluating the associated eigenvalues or characteristic equations at the equilibrium. The Jacobian matrix consists of the partial derivatives of each system equation with respect to the state variables. To determine the stability of an equilibrium point E , its characteristic equation is derived from the Jacobian matrix. By linearizing system (19), the perturbation dynamics can be expressed as $\dot{x} = AX$. The Jacobian matrix J is given is obtained by evaluating the partial derivatives of the system with respect to each state variable at the equilibrium point. For any equilibrium E , the corresponding Jacobian matrix J can therefore be calculated as follows.

$$J = \begin{bmatrix} -1 & 0.4 & -4 & 0 & 0 \\ -0.23 & -1 - 0.3 - kW & kW & 0 & 0 \\ 0 & -4.5 + kW & -1 - kW & 0 & 0 \\ 0 & 0 & 0 & -c & 0 \\ 0 & p & -p & 0 & qh' \end{bmatrix}. \quad (22)$$

where

$$W = \alpha - \beta\tilde{\varphi}_1 - \gamma\tilde{\varphi}_2. \quad (23)$$

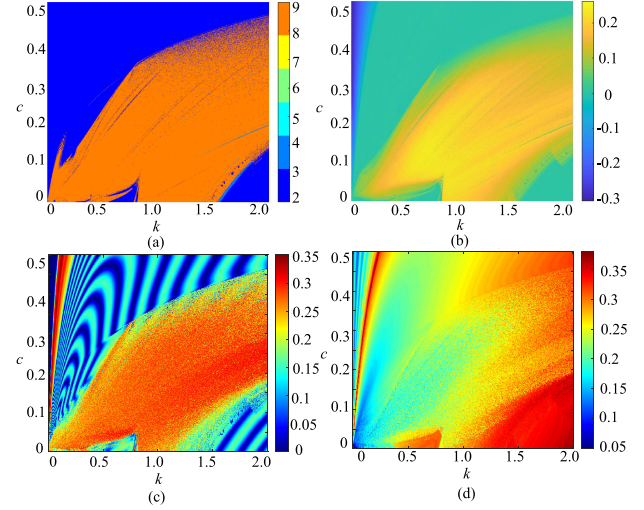


Fig. 6. The distribution of dynamic characteristics on the k - c plane. (a) 2D bifurcation diagram. (b) 2D maximum Lyapunov exponent graph. (c) 2D spectral entropy. (d) 2D approximate Entropy.

and

$$h' = \begin{cases} 1, M < 0 \\ 1 - s \sum_{i=0}^N \text{sech}^2(s(\tilde{\varphi}_2 + 2i)), M = 0, 1, 2, \dots \end{cases} \quad (24)$$

In Eq.(24), the derivative of the sign function is approximated using a hyperbolic tangent function, i.e., $\text{sign}(x) \approx \tanh(sx)$, where $s=10^6$. It can be observed that the Jacobian matrix depends solely on the system parameters $\alpha, \beta, \gamma, k, c, p, q$, and M . For a given parameter set $\alpha=\beta=1, \gamma=0.01, k=1, c=0.1, p=2.2$, and $q=5$, the stability properties of the equilibrium points under different cases are obtained through numerical simulations in MATLAB, as summarized in Table I. As indicated in Table I, the equilibrium points in Cases 1-4 correspond to a stable focus, an unstable saddle, a stable focus, and an unstable saddle, respectively. These results demonstrate that the SOM-HNN is capable of generating self-excited chaotic attractors.

C. Parameter-Controlled Multi-Butterfly and Multi-Scroll Attractors

In this subsection, the emergence of complex multi-butterfly and multi-scroll attractors in the constructed SOM-HNN is demonstrated. First, the control parameter $M < 0$ is investigated, i.e., $h = \varphi$. With parameters fixed as $\alpha = \beta = 1, \gamma = 0.01, a = 1, b = 1, p = 2.2, q = 5$, and initial values $(x_{10}, x_{20}, x_{30}, \varphi_{10}, \varphi_{20}) = (0.1, 0.1, 0.1, 0.1, 0.1)$, the memristor

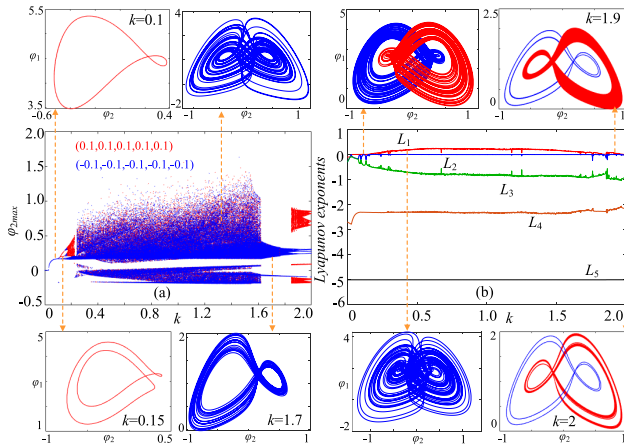


Fig. 7. k -dependent dynamical characteristics under $c = 0.1$. (a) Bifurcation diagram with different initial states, $(0.1, 0.1, 0.1, 0.1, 0.1)$ colored red and $(-0.1, -0.1, -0.1, -0.1, -0.1)$ colored blue. (b) Lyapunov exponents with initial states $(0.1, 0.1, 0.1, 0.1, 0.1)$.

coupling strength $k \in [0, 2]$ and parameter $c \in [0, 0.5]$ are chosen as bifurcation parameters. Using ODE23 solver, a 2-dimensional (2D) bifurcation diagram on the (k, c) plane is obtained, as shown in Fig. 6(a). Different colors indicate distinct dynamical regimes: orange for chaos, yellow for quasi-periodicity, green and cyan for multi-periodicity, light blue for period-2, and dark blue for period-1. This demonstrates that the proposed SOM-HNN exhibits a wide range of dynamics, including chaos, quasi-periodic, and periodic behaviors. The corresponding 2D maximum Lyapunov exponent diagram is presented in Fig. 6(b). It is not difficult to see that the LE diagram aligns with the bifurcation diagram. Furthermore, the two-dimensional spectral entropy (SE) and approximate entropy (AE) distributions are presented in Fig. 6(c) and (d), respectively. The results demonstrate that the generated chaotic sequences exhibit superior performance metrics, highlighting the applicability of the SO-MNN to a wide range of chaos-based applications. These findings further confirm that the introduction of the second-order memristor significantly enhances the chaotic complexity of the original Hopfield neural network.

To provide deeper insights, a k -dependent bifurcation diagram and the corresponding Lyapunov exponents are plotted in Fig. 7(a) and (b), respectively, with $c = 0.1$. Two sets of initial conditions, $(0.1, 0.1, 0.1, 0.1, 0.1)$ and $(-0.1, -0.1, -0.1, -0.1, -0.1)$, yield bifurcation curves colored red and blue, respectively. As k increases from 0 to 2, the dynamical behavior of the SOM-HNN transitions from a periodic state through a forward period-doubling bifurcation route into multi-periodic and quasi-periodic regimes. At $k = 0.3$, the system enters a robust chaotic state that persists until $k = 1.62$, beyond which chaos collapses into periodicity through a reverse period-doubling bifurcation route. Notably, multiple coexisting attractors emerge within intervals such as $k \in [0.11, 0.15]$, $k \in [0.16, 0.22]$, and $k \in [1.85, 2]$. Meanwhile, representative attractors obtained for different k values are illustrated in Fig. 7, including period-1 attractor ($k = 0.1$),

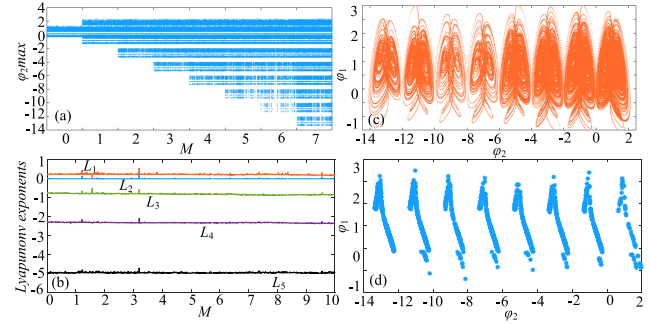


Fig. 8. Dynamics of the MHNN for the control parameter M . (a) bifurcation diagram. (b) Corresponding Lyapunov exponents. (c) 8-butterfly attractor phase portrait. (d) Corresponding Poincaré map.

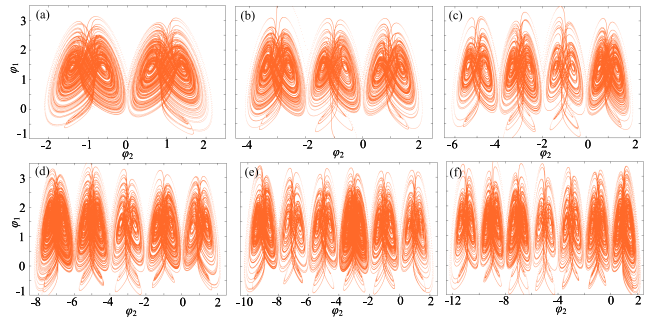


Fig. 9. For $k = 0.4$, multi-butterfly attractors of the MHNN with different values of M . (a) 2-butterfly attractor with $M = 0$. (b) 3-butterfly attractor with $M = 1$. (c) 4-butterfly attractor with $M = 2$. (d) 5-butterfly attractor with $M = 3$. (e) 6-butterfly attractor with $M = 4$. (f) 7-butterfly attractor with $M = 5$.

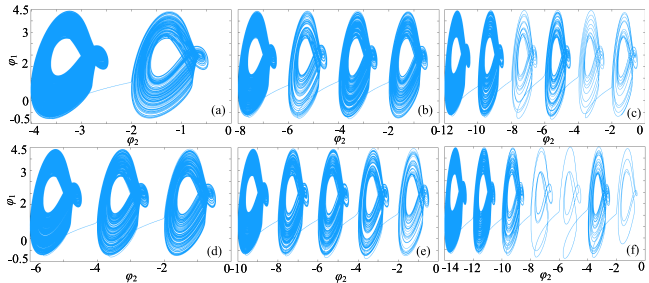


Fig. 10. For $k = 0.225$, Similar Multi-scroll attractors of the MHNN with different values of M . (a) 2-scroll attractor with $M = 1$. (b) 4-scroll attractor with $M = 3$. (c) 6-scroll attractor with $M = 5$. (d) 3-scroll attractor with $M = 2$. (e) 5-scroll attractor with $M = 4$. (f) 7-scroll attractor with $M = 6$.

period-2 attractor ($k=0.15$), coexisting symmetrical chaotic scroll attractors ($k = 0.2$), chaotic butterfly attractors ($k = 0.4$, $k = 1.5$), quasi-period attractors ($k = 1.7$), coexisting period and quasi-period attractors ($k = 1.9$), and coexisting symmetrical periodic attractors ($k = 2$). These results confirm that the proposed SOM-HNN exhibits remarkably rich dynamics, particularly chaotic scroll and butterfly attractors, which are rarely reported in neural network models.

Next, the control parameter M is treated as an adjustable variable, while keeping all other parameters unchanged except $k = 0.4$, $c = 0.1$, and initial states $(0.1, 0.1, 0.1, 0.1, 0.1)$.

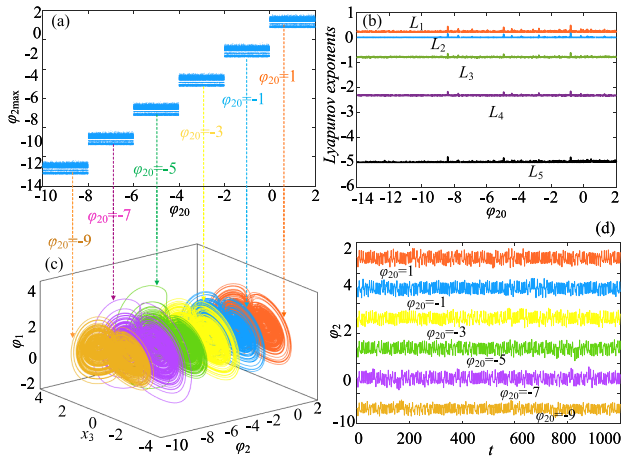


Fig. 11. Dynamics of the MHNN for the memristor initial value φ_{20} . (a) Bifurcation diagram. (b) Corresponding Lyapunov exponents. (c) Coexisting 6 butterfly attractors on $\varphi_2 - \varphi_3 - \varphi_1$ space. (d) Coexisting 6 chaotic sequences along φ_2 -direction.

As shown in the bifurcation diagram Fig. 8(a) and Lyapunov exponents Fig. 8(b), as M increases from 0 to 7, the SOM-HNN exhibits a stepwise bifurcation pattern along the φ_2 -direction, indicating that the chaotic butterfly and scroll attractor is expanded in this dimension. To validate this phenomenon, a complex 8-butterfly attractor is generated when $M = 6$, as illustrated in Fig. 8(c). Meanwhile, the corresponding Poincaré map, plotted under $x_2 = -0.1$ in Fig. 8(d), displays multiple irregular bars, confirming the formation of a multi-butterfly attractor with an extremely complicated topological structure. Moreover, to explore the relationship between the number of multi-butterfly attractors and the control parameter M , phase portraits under different values of M are presented in Fig. 9. The results demonstrate that the SOM-HNN can generate an arbitrary number of multi-butterfly attractors, with the count determined by $(M + 2)$. In addition, when $k = 0.225$, the SOM-HNN produces multi-scroll attractors under varying M , as shown in Fig. 10, where the number of scrolls is directly controlled by $(M + 1)$. These findings highlight a remarkable property of the proposed SOM-HNN: it can simultaneously generate two distinct categories of multi-structured attractors, multi-butterfly and multi-scroll, which has rarely been observed in previous neural network models.

D. Initial-Offset-Controlled Butterfly and Scroll Attractors

It is well known that biological nervous systems exhibit extreme sensitivity to their initial states. A similar phenomenon can be observed in the constructed SOM-HNN. Keeping the previous parameters unchanged except for $k = 0.4$, $p = 1$, $M = 4$, and initial states set as $(0.1, 0.1, 0.1, 0.1, \varphi_{20})$, the bifurcation diagram of state variable φ_2 and the first five finite-time Lyapunov exponent spectra are shown in Figs. 11(a) and (b), respectively, with $\varphi_{20} \in [-10, 2]$. As shown in Fig. 11(a), the bifurcation diagram exhibits multiple long strip-shaped regions of identical dynamical amplitude but different positions, each of which can evolve

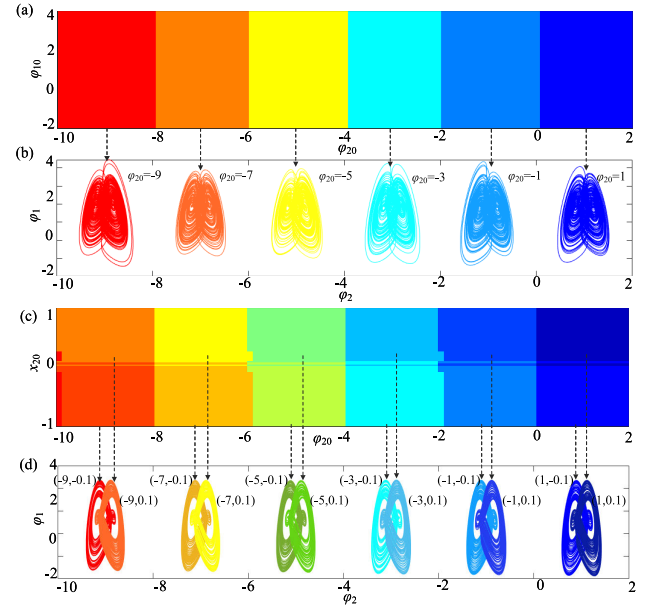


Fig. 12. The initial offset boosting behavior. (a) Attraction basin on $\varphi_{20} - \varphi_{10}$ plane with $k = 0.4$. (b) Coexisting six butterfly attractors with different positions. (c) Attraction basin on $\varphi_{20} - x_{20}$ plane with $k = 0.225$. (d) Coexisting twelve symmetrical scroll attractors with different positions.

into a butterfly attractor. Meanwhile, Fig. 11(b) reveals that the Lyapunov exponents remain invariant, indicating that the SOM-HNN is capable of generating continuously robust chaos. This demonstrates the existence of a complex initial offset boosting behavior, highly dependent on the initial states. To further confirm this property, phase portraits and corresponding time series are plotted in Figs. 11(c) and (d) for $\varphi_{20} = 1, -1, -3, -5, -7$, and -9 . As illustrated in Fig. 11(c), six coexisting butterfly attractors are symmetrically distributed along the φ_2 -axis, sharing the same structure but located at different positions, which directly correspond to the bifurcation structure in Fig. 11(a). For simplicity, only a subset of coexisting attractors is displayed. In fact, the number of coexisting butterfly attractors can be flexibly adjusted by selecting the control parameter M , while their positions can be tuned by varying the initial value φ_{20} . This clearly verifies the occurrence of initial offset boosting behavior. Moreover, as shown in Fig. 11(d), the chaotic sequences remain robust while their oscillation amplitudes can be nondestructively controlled by the initial value φ_{20} . Such a property is particularly significant for practical applications, as it enables the generation of abundant, nondestructive, and robust chaotic sequences, which are highly desirable for chaos-based secure communication systems.

In addition, the initial-boosting coexisting behaviors of the SOM-HNN are further revealed through the analysis of attraction basins, defined as the domains of initial conditions. With the above parameters fixed, the initial states φ_{20} and φ_{10} are scanned over the ranges $[-10, 2]$ and $[-2, 4]$, respectively, and the corresponding attraction basin in the $\varphi_{20} - \varphi_{10}$ plane is depicted in Fig. 11(a), where distinct colors represent attractors located at different positions. By selecting $\varphi_{20} = 1, -1, -3, -5, -7$, and -9 , six

butterfly attractors with distinct positions are obtained, as shown in Fig. 11(b), demonstrating the occurrence of initial-boosting coexistence in the SOM-HNN. Furthermore, with $k = 0.225$, the attraction basin in the $\varphi_{20} - x_{20}$ plane for $\varphi_{20} \in [-10, 2]$ and $x_{20} \in [-1, 1]$ is illustrated in Fig. 12(c). Unlike Fig. 11(a), this attraction basin exhibits richer color diversity and symmetry about $x_{20} = 0$, indicating that the dynamics of the SOM-HNN depend not only on φ_{20} but also on x_{20} . By selecting six pairs of initial values $(\varphi_{20}, x_{20}) = (-9, \pm 0.1), (-7, \pm 0.1), (-5, \pm 0.1), (-3, \pm 0.1), (-1, \pm 0.1)$, and $(1, \pm 0.1)$, twelve coexisting symmetrical scroll attractors are generated, as shown in Fig. 12(d). These results demonstrate that the proposed SOM-HNN is capable of simultaneously producing two distinct forms of initial-boosting coexisting behaviors, thereby exhibiting highly complex dynamical properties.

IV. ANALOG CIRCUIT AND FPGA IMPLEMENTATION

With the rapid development of neuromorphic computing, hardware implementation of neural networks has become a critical research direction to achieve high efficiency, low latency, and energy-aware computation. Among the available approaches, analog circuits and field-programmable gate arrays (FPGAs) are two widely adopted platforms [45], [46]. Analog circuits provide inherent parallelism and high computational density while consuming very low power, making them suitable for real-time processing and edge intelligence. However, their design complexity, susceptibility to noise, and limited scalability impose significant challenges for large-scale neural network implementations. In contrast, FPGAs offer flexible reconfigurability, high throughput, and a balance between performance and programmability, which enable rapid prototyping and efficient mapping of complex neural architectures. Nevertheless, FPGA implementations generally require higher power consumption and larger hardware resources compared to analog counterparts, and they may struggle to match the intrinsic efficiency of analog designs in certain tasks. Therefore, exploring the implementation method of analog and FPGA-based neural networks provides valuable insights for selecting appropriate implementation strategies and optimizing neuromorphic hardware systems.

A. Analog Circuit Design and Multisim Simulation

In this subsection, an analog SOM-HNN circuit is designed, and its numerical results are validated through simulations in the Multisim. Based on the mathematical model in (19), the corresponding circuit equations of the analog SOM-HNN can be expressed as follows:

$$\begin{cases} C\dot{v}_{x1} = -v_{x1}/R + \tanh(v_{x2})/R_1 - \tanh(v_{x3})/R_2, \\ C\dot{v}_{x2} = -v_{x2}/R - \tanh(v_{x1})/R_3 - \tanh(v_{x2})/R_4 \\ \quad - g(1/R_k R_\alpha - v_{\varphi 1}/R_k R_\beta - v_{\varphi 2}/R_k R_\gamma)(v_{x2} - v_{x3}), \\ C\dot{v}_{x3} = -v_{x3}/R - \tanh(v_{x2})/R_5 \\ \quad + g(1/R_k R_\alpha - v_{\varphi 1}/R_k R_\beta - v_{\varphi 2}/R_k R_\gamma)(v_{x2} - v_{x3}), \\ C\dot{v}_{\varphi 1} = e/R_\alpha - g(v_{x2} - v_{x3})^2/R_\beta - v_{\varphi 1}/R_c, \\ C\dot{v}_{\varphi 2} = (v_{x2} - v_{x3})/R_p - h(v_{\varphi 2})/R_q, \end{cases} \quad (25)$$

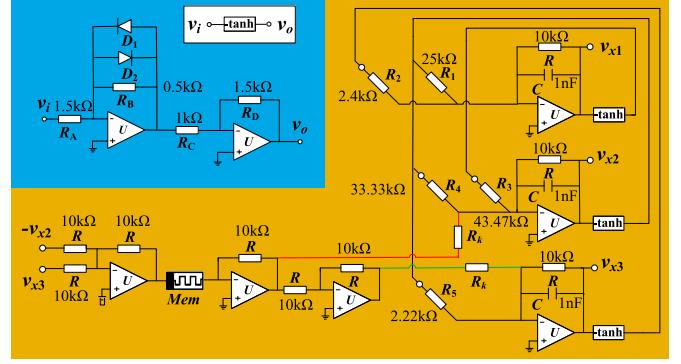


Fig. 13. Analog SOM-HNN circuit: $R_A = 2.54 \text{ k}\Omega$, $R_B = 0.54 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_D = 1.5 \text{ k}\Omega$, $R = 10 \text{ k}\Omega$, $R_1 = 25 \text{ k}\Omega$, $R_2 = 2.4 \text{ k}\Omega$, $R_3 = 43.47 \text{ k}\Omega$, $R_4 = 33.33 \text{ k}\Omega$, $R_5 = 2.22 \text{ k}\Omega$, $C = 1 \text{ nF}$, and R_k is adjustable resistor.

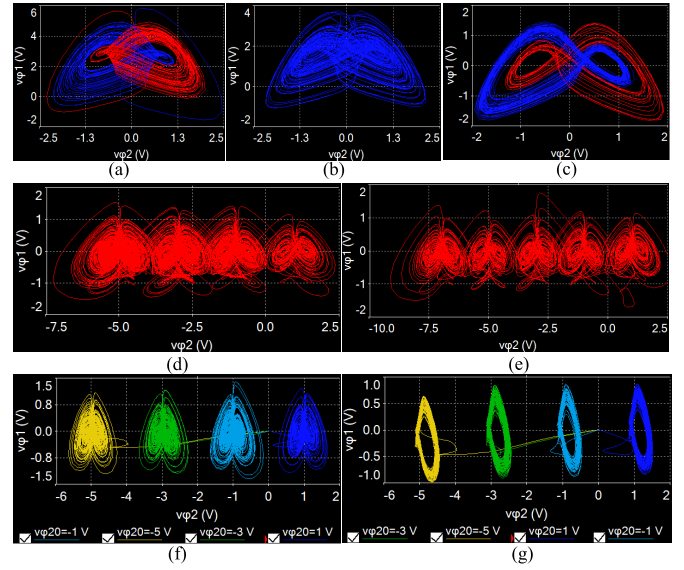


Fig. 14. Circuit experimental results captured from the MHHN circuit. (a) Coexisting symmetrical chaotic attractors with $R_p = 4.54 \text{ k}\Omega$, $R_k = 50 \text{ k}\Omega$ (correspond to Fig. 7 under $k = 0.2$). (b) Chaotic butterfly attractor with $R_p = 4.54 \text{ k}\Omega$, $R_k = 25 \text{ k}\Omega$ (correspond to Fig. 7 under $k = 0.4$). (c) Coexisting symmetrical quasi-period attractors with $R_p = 4.54 \text{ k}\Omega$, $R_k = 5.2 \text{ k}\Omega$ (correspond to Fig. 7 under $k = 1.9$). (d) 4-butterfly attractor with $R_k = 25 \text{ k}\Omega$ (correspond to Fig. 9(c)). (e) 5-butterfly attractor with $R_k = 25 \text{ k}\Omega$ (correspond to Fig. 9(d)). (f) Coexisting 4 butterfly attractors with $R_p = 10 \text{ k}\Omega$, $R_k = 25 \text{ k}\Omega$ (correspond to Fig. 12(b)). (g) Coexisting 4 scroll attractors with $R_p = 10 \text{ k}\Omega$, $R_k = 45 \text{ k}\Omega$ (correspond to Fig. 12(d)).

where RC is the integral time constant, the capacitor voltages v_{x1} , v_{x2} , v_{x3} , $v_{\varphi 1}$, and $v_{\varphi 2}$ corresponded to the five state variables x_1 , x_2 , x_3 , φ_1 and φ_2 , respectively. Based on the circuit equations in (10) and the second-order memristor circuit shown in Fig. 13, an analog SOM-HNN circuit is designed as illustrated in Fig. 13, where the hyperbolic tangent function circuit reported in [11] is adopted. Resistors $R_1 - R_5$ represent fixed synaptic weight coefficients, while the effect of magnetic coupling in the neural network is simulated by the second-order memristor Mem. The coupling coefficient k is realized using adjustable resistor R_k , defined as $R_k = R/k$.

The designed SOM-HNN circuit emulator is implemented in the Multisim simulation soft using commercially available components such as M/AD633JN, U/TL082CP, and a $\pm 15 \text{ V}$

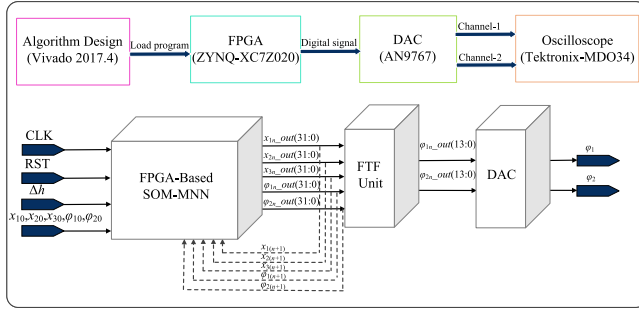


Fig. 15. Process of FPGA-based hardware implementation.

DC power supply. By setting $C=1$ nF and $R=10$ k Ω , the resistors corresponding to the fixed synaptic weight coefficients are determined as $R_1=25$ k Ω , $R_2=2.4$ k Ω , $R_3=43.47$ k Ω , $R_4=33.33$ k Ω , and $R_5=2.22$ k Ω . By adjusting the resistance of R_k , applying the appropriate input voltages e_i , and switching S1-SM, the analog SOM-HNN circuit reproduces the behaviors observed in the numerical simulations. For instance, in comparison with the simulation results in Fig. 7, the circuit generates consistent attractors when $k = 0.2, 0.4$, and 1.9 , corresponding to $R_k=50$ k Ω , 25 k Ω , and 5.2 k Ω , respectively, as shown in Fig. 14(a)-(c). Similarly, for the results in Fig. 9(c)-(d), with $p = 2.2$ and $k = 0.4$, the resistor values are set as $R_p=4.45$ k Ω and $R_k=25$ k Ω . By closing switches S1-S2 and S1-S3, the circuit produces 4- and 5-butterfly attractor, as illustrated in Fig. 14(d)-(e). Additionally, under $p = 1$ and $k = 0.4$ or 0.225 , the resistor values are configured as $R_p=10$ k Ω , $R_k=25$ k Ω , and 45 k Ω , respectively. By closing switches S1-S2 and selecting different initial capacitor voltages $v\phi_{20}$, the circuit yields coexisting four butterfly attractors and coexisting four scroll attractors as shown in Fig. 14(g)-(f), consistent with the simulation results in Fig. 12(b)-(d). These experimental observations confirm that the designed analog SOM-HNN circuit reliably reproduces the numerical simulation results, thereby validating its physical realizability and robustness.

B. Digital Circuit Design and Hardware Measurement

In this subsection, the SOM-HNN is realized and experimentally verified on an FPGA platform. The schematic diagram of the FPGA-based hardware setup is illustrated in the upper section of Fig. 15. The experimental environment employs the Vivado 2017.4 design suite as the development tool, while the AX7020 board equipped with a Xilinx ZYNQ-XC7Z020 device serves as the hardware platform. The digital-to-analog conversion is performed using the AN9767 module, and signal observation is carried out with a Tektronix MDO34 dual-channel oscilloscope.

The designed SOM-HNN is discretized and implemented in Vivado using Verilog HDL, where the fourth-order Runge-Kutta (RK4) method is adopted for numerical computation. The floating-point arithmetic complies with the IEEE 754 standard, and the discrete time step Δh is set to 0.01. The lower part of Fig. 15 presents the primary workflow of the FPGA realization. As depicted, the chaotic oscillator block

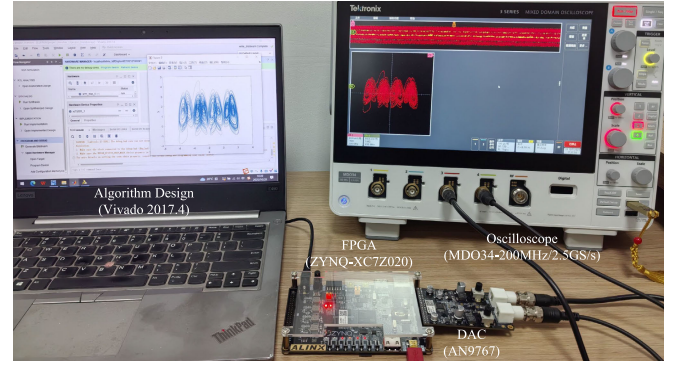


Fig. 16. Platform of FPGA-based hardware implementation.

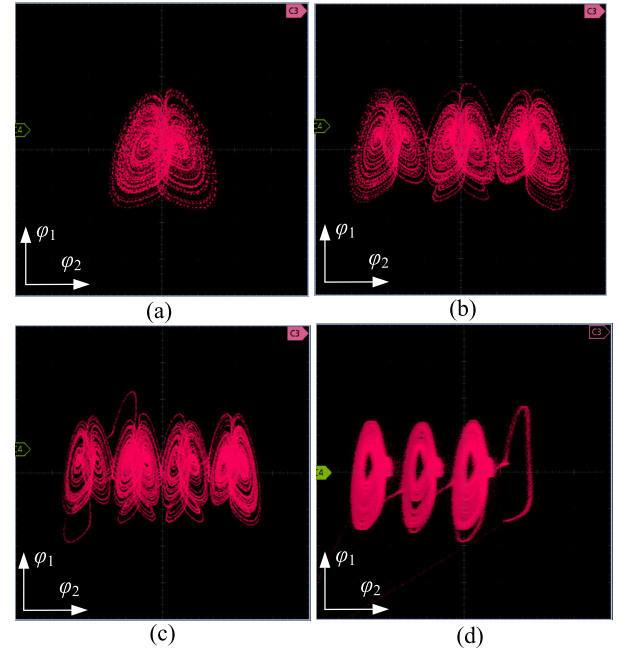


Fig. 17. FPGA-based experiment results. (a) 1-butterfly attractor. (b) 3-butterfly attractor. (c) 4-butterfly attractor. (d) 3-scroll attractor.

primarily applies the RK4 scheme to compute the dynamic behavior of the neural network. It consists of four input and four output ports. The system clock “CLK” and reset signal “RST” act as 1-bit control inputs, while Δh and the initial conditions $(x_{10}, x_{20}, x_{30}, \phi_{10}, \phi_{20})$ are provided as 32-bit data inputs. Correspondingly, the five 32-bit output parameters $(x_{1out}, x_{2out}, x_{3out}, \phi_{1out}, \phi_{2out})$ are fed back as initial states for the next iteration of the oscillator. Simultaneously, these outputs are sent to the FTF (floating-to-fixed) conversion unit, where they are transformed into 14-bit fixed-point representations. Considering the phase trajectories on the $\phi_1 - \phi_2$ plane and aiming to economize hardware resources, only two 14-bit output channels (ϕ_{1out} and ϕ_{2out}) are finally retained for implementation.

The FPGA-based experimental setup for realizing the SOM-HNN is illustrated in Fig. 16. The oscilloscope displays the analog voltage outputs from two separate channels. As depicted in Fig. 17, by adjusting various control parameters, several representative multi-butterfly and multi-scroll

TABLE II
STATISTICAL RESULTS OF FPGA RESOURCE CONSUMPTION
AND UTILIZATION

Resource	Available	Used	Utilization
LUT	53200	30362	57.07 %
FF	106400	2085	1.96 %
BRAM	140	0	0
DSP	220	25	11.36 %
IO	200	70	35 %
BUFG	32	2	6.25 %

TABLE III
FPGA HARDWARE PERFORMANCE ANALYSIS

Maximum frequency	Throughput	Latency	Total static power	Total power
127.36 MHz	65.34 Mbit/s	4.6 ms	82 mW	268 mW

attractors emerge on the oscilloscope screen. These experimentally observed patterns closely correspond to the numerical simulations presented in Section III. Table II summarizes the hardware resource utilization of the proposed FPGA-based implementation, where the optimal results in terms of LUT, FF, BRAM, DSP, IO, and BUFG usage are highlighted. The experimental results demonstrate that the proposed design achieves a favorable balance between functionality and resource efficiency, without imposing excessive hardware overhead. Specifically, the utilization ratios of LUT and FF are 57.07 % and 1.96 %, respectively, indicating moderate combinational logic usage and extremely low sequential logic consumption. Notably, no BRAM resources are required, which significantly enhances the scalability and portability of the design. In addition, the usage of DSP, IO, and BUFG resources remains limited to 11.36 %, 35 %, and 6.25 %, respectively, further confirming the hardware-lightweight nature of the implementation. Table III presents the corresponding hardware performance metrics, with the best results highlighted in terms of maximum operating frequency, throughput, latency, and power consumption. As shown, the proposed architecture operates at a maximum frequency of 127.36 MHz, enabling a high data throughput of 65.34 Mbit/s while maintaining a low processing latency of 4.6 ms. Moreover, the power analysis reveals excellent energy efficiency, with a total static power consumption of only 82 mW and a total power consumption of 268 mW. These results collectively indicate that the proposed FPGA-based SOM-HNN implementation not only delivers high throughput and low latency but also exhibits low power consumption, making it particularly suitable for high-performance and resource-constrained applications.

V. CONCLUSION

In this work, a novel SOM-based approach has been developed to enhance chaos in neural networks. By integrating a multi-piecewise second-order memristor into a simple three-neuron Hopfield neural network to model the magnetic coupling effect, a SOM-HNN is proposed. The system

exhibits highly diverse and complex dynamics. Both multi-butterfly and multi-scroll attractors are generated through memristor-related control parameters, while infinitely many coexisting single-butterfly and single-scroll attractors emerge under memristor-related initial conditions. These phenomena reveal extremely rich and complex dynamical behaviors. Unlike existing studies, the proposed SOM method significantly improves both the diversity and complexity of chaotic dynamics, thereby enhancing the overall performance of neural networks. To further demonstrate its practical potential in neuromorphic systems, an analog SOM-HNN circuit was designed and a digital implementation was realized on FPGA. Experimental results are shown to be consistent with numerical simulations, validating the feasibility of the proposed model.

Future research will focus on extending this approach to design and optimize neural networks with larger scales and to explore their integration into applications leveraging chaotic dynamics.

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Hairong Lin (Senior Member, IEEE) received the M.S. and Ph.D. degrees in information and communication engineering and computer science and technology from Hunan University, Changsha, China, in 2015 and 2021, respectively. From 2022 to 2023, he was a Post-Doctoral Fellow with the School of Computer Science and Electronic Engineering, Hunan University. He is currently an Associate Professor with the School of Electronic Information, Central South University, Changsha. He has presided over three national and provincial projects and published more than 50 articles in related international journals, such as *IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS*, *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS*, *IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS*, and *IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS*. His research interests include memristive neural networks, chaotic systems and circuits, information and network security, and the Internet of Things. He is a member of the Chaos and Nonlinear Circuit Professional Committee of the Circuit and System Branch of China Electronic Society.



Xiaoheng Deng (Senior Member, IEEE) received the Ph.D. degree in computer science from Central South University, Changsha, Hunan, China, in 2005. Since 2006, he has been an Associate Professor and then a Full Professor with the Department of Communication Engineering, Central South University. He is a joint Professor with Shenzhen Research Institute, Central South University, and the Director of the Data Sensing and Switching Equipment Provincial Engineering Center. His research interests include edge computing, the Internet of Things, wireless networking and communication, data mining, and pattern recognition. He is a Senior Member of CCF and a member of the CCF Pervasive Computing Council and ACM. He has been the Chair of CCF YOCSEF CHANGSHA from 2009 to 2010.



Geyong Min (Member, IEEE) received the B.Sc. degree in computer science from the Huazhong University of Science and Technology, China, in 1995, and the Ph.D. degree in computing science from the University of Glasgow, U.K., in 2003. He is currently a Professor of high-performance computing and networking with the Department of Computer Science, University of Exeter, U.K. His research interests include computer networks, wireless communications, parallel and distributed computing, ubiquitous computing, multimedia systems, modeling, and performance engineering.



Yi Zhang received the bachelor's degree in communication engineering from the Computer and Information Engineering College, Three Gorges University. He is currently pursuing the M.S. degree with Central South University. His research interests include memristive neural networks, chaotic systems and circuits, brain-like computing, and privacy protection.